6-4. Multiple regression: interpretation

- Some caution needed when interpreting multiple regression analysis.
- Consider importance/effect of an explanatory variable *in the presence of other variables* already in the model.
- The coefficients/parameters and associated standard errors and p-values do not tell the whole story.
- Need to weigh up all the evidence – all the plots and summary statistics, including the contributions to the total Sum of Squares and the overall coefficient of determination R^2

Example 6.3

Return to data of example 6.1 and fit only one explanatory variable. First fit x_1 alone.

```
> model1 = lm(Y ~ x1)
> summary(model1)
Call:
lm(formula = Y ~ x1)

Residuals:
      1       2       3       4       5       6       7
 0.4287 -0.1690  0.0279 -0.2705  0.1349 -0.2737  0.1217

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.004506   0.683082  -0.007  0.99499
x1           0.992201   0.218681   4.537  0.00618 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2808 on 5 degrees of freedom
Multiple R-squared:  0.8046,    Adjusted R-squared:  0.7655
F-statistic: 20.59 on 1 and 5 DF,  p-value: 0.006185
```

Coefficient β_1 is highly significant, with p-value 0.00618.

$R^2 = 0.8046$, variable x_1 explains 80.5% of the variation.

```
> aov(model1)

Call:
   aov(formula = model1)

Terms:
                       x1 Residuals
Sum of Squares  1.6229574 0.3941854
Deg. of Freedom         1         5

Residual standard error: 0.2807794
Estimated effects may be unbalanced
```

Total variation is $S_{yy} = 1.623 + 0.394 = 2.017$.

Variable x_1 accounts for 1.623 of this, which is 80.5% of the total.

Recall original analysis, with both x_1 and x_2

```
> regression = lm(Y ~ x1 + x2)
> summary(regression)

Call:
lm(formula = Y ~ x1 + x2)

Residuals:
       1        2        3        4        5        6        7
 0.40509 -0.07718  0.16946 -0.28475  0.01181 -0.23812  0.01368

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.21382    0.75024  -0.285   0.7898
x1           0.89843    0.25300   3.551   0.0238 *
x2           0.01745    0.02116   0.825   0.4558
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2902 on 4 degrees of freedom
Multiple R-squared:  0.833,     Adjusted R-squared:  0.7495
F-statistic: 9.975 on 2 and 4 DF,  p-value: 0.02789
```

Together the variables explain 83.3% of the variation (not much more than x_1 alone).

```
> aov(regression)
Call:
   aov(formula = regression)

Terms:
                        x1        x2 Residuals
Sum of Squares  1.6229574 0.0573038 0.3368816
Deg. of Freedom         1         1         4

Residual standard error: 0.2902075
Estimated effects may be unbalanced
```

In the presence of x_1, adding x_2 accounts for only a further 0.0573 of the total variation.

Now try fitting x_2 alone.

```
> model2 = lm(Y ~ x2)
> summary(model2)
Call:
lm(formula = Y ~ x2)

Residuals:
      1       2       3       4       5       6       7
 0.3697  0.3258  0.3818 -0.2303  0.3576 -0.3742 -0.8303

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.59394    1.00446   1.587    0.173
x2           0.05121    0.03445   1.486    0.197

Residual standard error: 0.5289 on 5 degrees of freedom
Multiple R-squared:  0.3065,    Adjusted R-squared:  0.1678
F-statistic:  2.21 on 1 and 5 DF,  p-value: 0.1973
```

```
> aov(model2)

Call:
   aov(formula = model2)

Terms:
                        x2 Residuals
Sum of Squares  0.6182035 1.3989394
Deg. of Freedom         1         5

Residual standard error: 0.5289498
Estimated effects may be unbalanced
```

Total variation is $S_{yy} = 0.618 + 1.399 = 2.017$, as before.

Variable x_2 accounts for 0.618 of this, which is 30.6% of the total.

Coefficient β_2 is not significantly different from zero (even at 10% level), with p-value 0.197.

What happens if we fit x_2 first, then x_1?

```
> model21 = lm(Y ~ x2 + x1)
> summary(model21)
Call:
lm(formula = Y ~ x2 + x1)

Residuals:
       1        2        3        4        5        6        7
 0.40509 -0.07718  0.16946 -0.28475  0.01181 -0.23812  0.01368

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.21382    0.75024  -0.285   0.7898
x2           0.01745    0.02116   0.825   0.4558
x1           0.89843    0.25300   3.551   0.0238 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2902 on 4 degrees of freedom
Multiple R-squared:  0.833,     Adjusted R-squared:  0.7495
F-statistic: 9.975 on 2 and 4 DF,  p-value: 0.02789
```

```
> aov(model21)

Call:
   aov(formula = model21)

Terms:
                       x2        x1 Residuals
Sum of Squares  0.6182035 1.0620578 0.3368816
Deg. of Freedom         1         1         4

Residual standard error: 0.2902075
Estimated effects may be unbalanced
```

The variation explained by x_2 here is 0.618, the same as when only x_2 was fitted.

Once x_2 has been fitted, the remaining variation explained by x_1 is 1.062, *not* the same as when only x_1 was fitted.

Together the two variables explain 83.3% of the variation, as before.

Multiple regression: Interpretation

- The **order** in which variables are fitted **does make a difference** in how much variation each explains.
- `summary` t-tests indicate *marginal* significance of the variable in question, given the presence of *all* the other variables.
- `anova` F-tests are sequential — they indicate the significance of the variable in question, given the presence of those variables that appear *above* it in the `anova` output.
- It is clearer and simpler if we fit the variable which explains most of the variation in the responses first, then the variable of next importance, and so on, as measured by SS values.